CSE/ECE 538
Wireless Networks

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Overview

- Core elective course: open to PG and UG students
- Prerequisites:
  - **Mandatory**
    - Computer Networks (CSE 232)
    - Introduction to Programming (CSE 101)
  - **Desirable**
    - Operating systems (CSE 231)
    - Advanced programming (CSE 201)
- Topics in wireless communications and mobile systems

Overview

- Core elective course: open to PG and UG students
- Topics in wireless communications and mobile systems
- Wireless LANs, cellular systems, sensor networks etc.

Grading

- Assignment 25%
- Course project 25%
- Short in-class quizzes 10%
- Mid-semester-15%
- End semester-25%

Course outline

- Overview of wireless and mobile systems (wireless LANs, cellular systems, sensor networks, etc.) and the challenges therein.
- Wireless Physical layer design
- Wireless link layer: Medium access protocol, multiplexing, link adaptation
- Link layer multicast
- Energy Security concern

Course outline (2)

- Next generation wireless networks: IEEE 802.11ax, IEEE 802.11be
- Network layer: mobility management, cellular handoff, Mobile IP
- Multihop routing protocol
- TCP over wireless, other transport layer issues
- Multipath TCP
- Edge computing and its communication perspective
- Future directions

Course project

- First hand experience in doing simple wireless networking research
- A set of topics will be floated, you need to choose one amongst them or choose on your own
- We will analyze and understand wireless network performance in a real life scenario, probably using a combination of simulations and analysis of real traces.
- You might use a network simulator (NS-3) to simulate a WiFi network. You will run various simulations to understand WiFi performance at various layers (link layer, TCP, HTTP etc.).

Honor code

- No Copying
- F grade on first offense of copying.
- Semester expulsion on second offense
- Further offense no less than one year of expulsion

Readings

- Slides from lectures will be available on the course web page.
- Reference text book: Wireless Communication by ANDREA GOLDSMITH and Mobile Communications (2nd ed.). Jochen Schiller.
- Research papers will also serve as references

Course webpage

- Register yourself at Classcode

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